

Code-Layout, Readability and Reusability

Date	14 July 2026
Team ID	SWTID-2026-2280
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

Code Quality Parameter	Followed	Remarks
Consistent Indentation	Yes	Uniform formatting
Proper File Structure	Yes	Separate frontend and backend folders
Meaningful Variable Names	Yes	Easy to understand
Function / Method Names	Yes	Descriptive names
Code Comments	Yes	Important sections documented
Modular Design	Yes	APIs separated into routes
No Redundant Code	Yes	Reusable functions used
Error Handling	Yes	API exceptions handled

Reusable Components / Modules:

Component	Language / Technology	Where Reused	Reusability Level
Gemini Service	Python	All AI APIs	High
API Routes	FastAPI	Multiple Features	High
JavaScript Functions	JavaScript	UI interactions	Medium
Tailwind Components	HTML/CSS	Entire UI	High

Overall Code Quality Assessment:

Aspect	Rating	Comments
Code Layout & Structure	5	Well organized
Readability	5	Easy to understand
Reusability	5	Modular code design
Documentation / Comments	4	Sufficient documentation
Overall Score	4.8 / 5	Clean, maintainable and scalable code